

glm_analysis

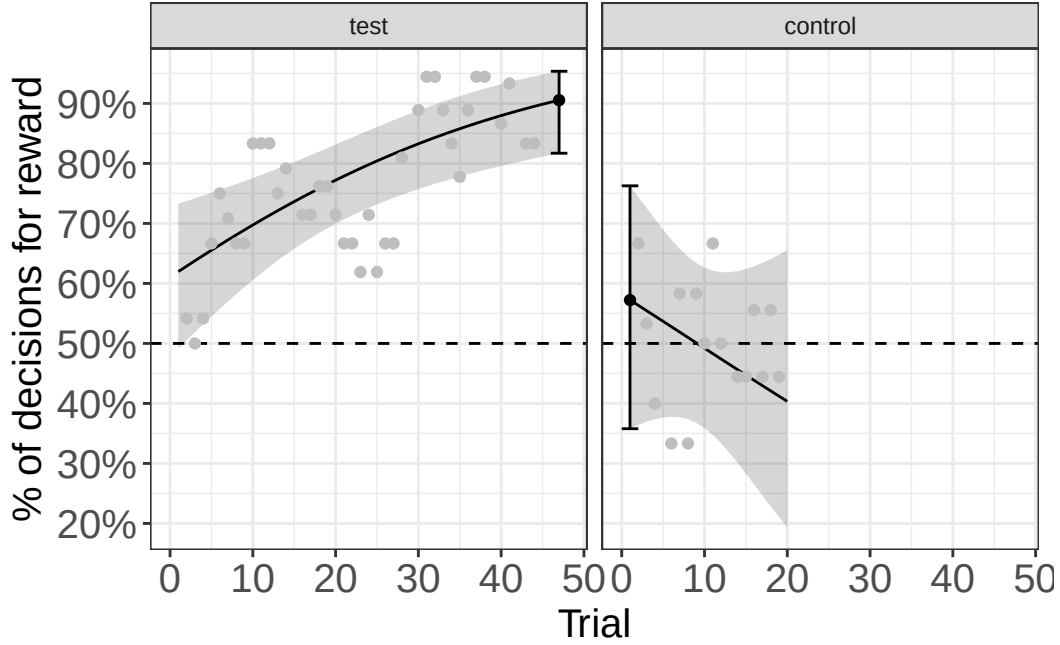
Make the computations necessary for all questions:

	contrast	odds.ratio	SE	df	null	z.ratio
1	test_end / control_start	2.161893	1.3957117	Inf	1	1.1942185
2	test_end / control_start	1.302833	0.5337640	Inf	1	0.6457023
3	test_end / control_start	2.921793	1.3532234	Inf	1	2.3150201
4	test_end / control_start	0.649945	0.3150206	Inf	1	-0.8889584
5	test_end / control_start	7.156515	3.9704981	Inf	1	3.5472092
6	test_end / control_start	7.059280	3.7730719	Inf	1	3.6565048

	p.value	experiment	control_start	test_end
1	0.2323924959	natcan1_170923	1	119
2	0.5184721949	natcan3_080825	1	64
3	0.0206118297	natcan3_190925	1	131
4	0.3740254126	natcan4_180825	1	100
5	0.0003893353	perpPara_170725	1	47
6	0.0002556776	thin0b_150823	1	66

Question 1: Can wasps use dorsal patterns?

- Looking at test versus control proportion correct
- Slope in test and control
- EMmeans at end of test and control
 - is this essentially equivalent to binning the last n trials and making the comparison



[1] "0.817-0.954"

		contrast	logOddsRatio	SE	df	asyp.LCL	asyp.UCL	z.ratio
1	test_end - control_start		1.968	0.555	Inf	0.881	3.055	3.547
	p.value	experiment	control_start	test_end	pFmt	confint		
1	0.000	perpPara_170725	1.000	47.000	<0.001	0.881-3.055		

During the test phase of the experiment, trial number (experience) had a significant positive effect on the expected probability (P_{corr}) of a correct decision (Figure 1 solid line) with every successive trial increasing the log odds of a correct decision by 0.043 (95% CI 0.019-0.067, $p < 0.001$).

The P_{corr} of a correct choice at the end of the test was higher ($P_{\text{corr}} = 0.905$, 95% CI 0.817-0.954) than at the start of the control ($P_{\text{corr}} = 0.572$, 95% CI: 0.358-0.763), and this difference was significant (log odds ratio = 1.968, $p < 0.001$) (Figure 1, compare error bars between test and control panel).

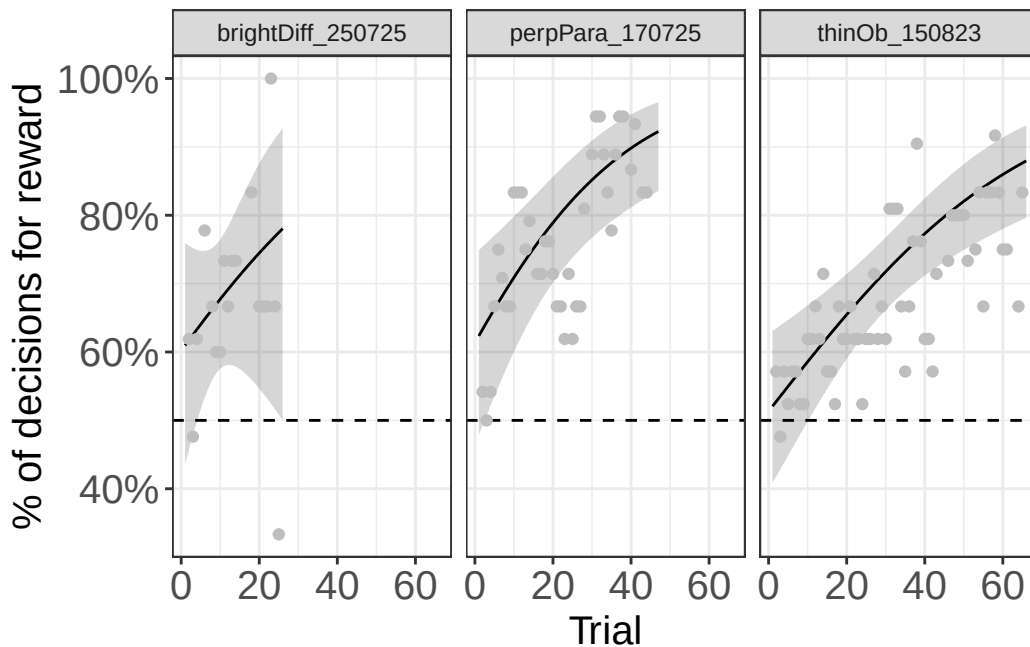
Also note that P_{corr} is significantly above chance (95% confidence ribbon does not overlap the 50% line) from an early trial number and remains that way until the end of the test. In contrast, the P_{corr} in the control phase is never significantly above chance for the duration of the control (95% confidence ribbon overlaps the 50% line).

That the model suggests naive wasps perform above chance at the onset of the test phase (ie. in trial 1) can be attributed to the fact that the model is unable to fit the rapid learning of

the wasps, rather than to the availability and learning of olfactory and / or temporal cues during the pre-test training phase. This is supported by the proportion of correct decisions in the first few 3-trial wide windows of the test phase (black points) that are clustered around 50%.

Question 2: What information are wasps using to discriminate patterns?

- Do they need optic flow?
- Can they distinguish simple brightness differences?
- are there differences in performance among artificial stimuli?



```
# A tibble: 3 x 5
  experiment      trialLogOdds ci95      pFmt oddsMultiplierPerTrial
  <chr>          <chr>      <chr>    <chr>    <chr>
1 brightDiff_250725 0.033    -0.037-0.103 0.356 1.033
2 perpPara_170725  0.043     0.019-0.067 <0.001 1.044
3 thinOb_150823    0.029     0.015-0.044 <0.001 1.030
```

To answer whether wasps require optic flow information to discriminate patterns we tested whether they could distinguish between oblique stripes oriented at 45 degrees to the right or left of the y-maze arm's long axis and the general direction of wasp travel.

Wasps had no trouble distinguishing between oblique stripes that presumably produce identical rates of optic flow in the dorsal visual field (Figure 2). Similar to the perpendicular-parallel experiment, trial had a significant positive effect on P_{corr} , with every successive trial increasing the log odds of a correct decision by xxx.

Initial performance (at trial 1) was not significantly different from chance ($P_{\text{corr}} = 0.521$, 95% CI:) but rose steeply, reaching 0.880 (95% CI:) by the end of the experiment (trial 66), which is significantly higher than expected by chance (50%).

Next, we asked whether wasps could discriminate between an opaque and transparent roof, ie. between stimuli that carried little to no pattern information. At the onset of the experiment (trial 1), wasp performance was not significantly different from 50% ($P_{\text{corr}} = 0.610$, 95% CI:)) but rose with experience and by the end of the experiment (trial 26) P_{corr} was 0.780 (95% CI:).

Performance at the end of the experiment is only borderline significantly higher than 50% because of the increased uncertainty in the slope / effect of trials on performance due to the lower number of trials run in this experiment. However, if one uses other metrics or analyses to compare performance to the 50% line, they are significantly different (for example, comparing performance at an intermediate trial or comparing the average performance of the last five trials to the 50% line).

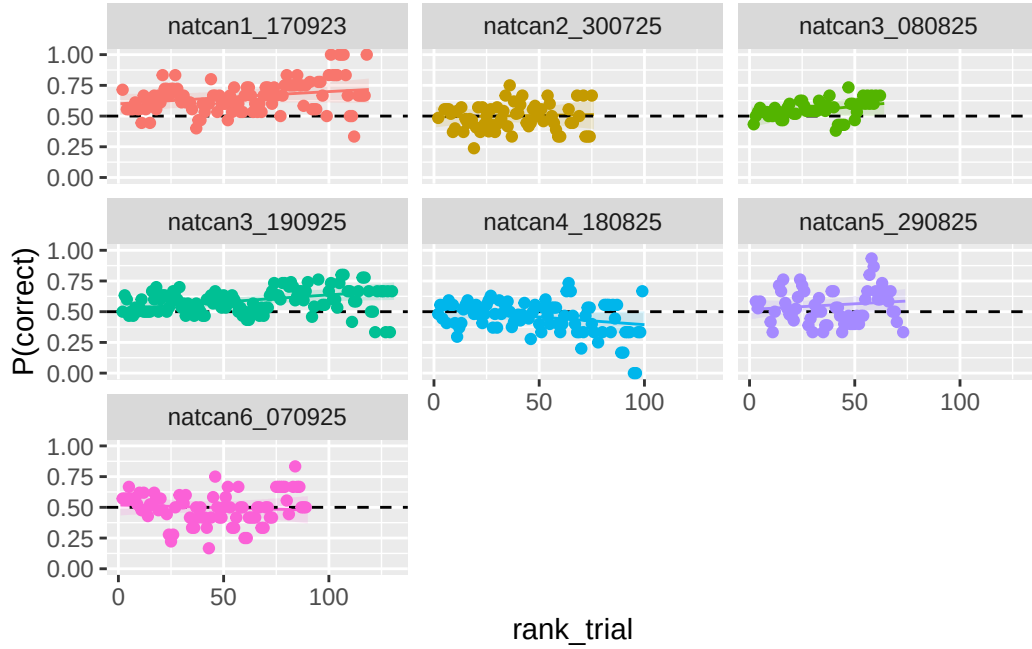
NBNB! Slope or brightdiff exp not significant. remove best fit line.

Comparison among different artificial stimuli:

issues: - day is multicollinear in thin oblique experiment. It makes the slope of thin oblique appear less steep. This drives the difference in slopes detected between perppara and thin oblique. The issue is, is it real? I'm not confident enough that I ran the experiments in exactly the same way so as to make them comparable. If we remove day from the analysis, would we still find a significant difference in the slopes? Could the difference simply be attributable to slower learning due to the side bias which is clearly evident in the thin oblique experiment (possibly caused because of different pre-testing training regimes). Why even is there a side bias when these wasps had just come off the back of a previous experiment with a thick oblique stripe? was the side bias present at that time? - the issue im trying to get at above is that conclusions regarding slope diffs depend on the model you run. - massive confidence intervals around brightness experiment

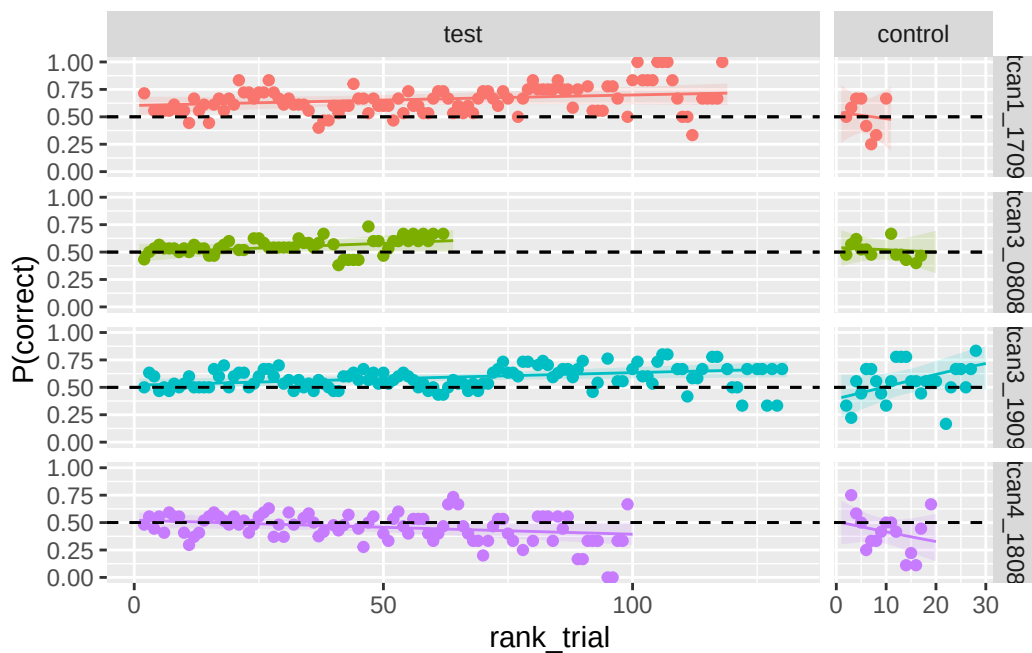
Question 3: Can wasps use naturalistic patterns for navigation?

- Are performances different from 50%?
- Are performances different from the control?
- Are there differences in performance between naturalistic and artificial stimuli?



Discrimination performance for naturalistic canopies tends to be lower than for artificial canopies (make some sort of comparison or just give ranges) and evidence for whether wasps can discriminate naturalistic canopies at above-chance rates ($>50\%$) is mixed. While P_{corr} was generally lower than for the artificial canopies, in four of the seven naturalistic canopy pairs we tested, the predicted P_{corr} by the end of the experiment was significantly above chance (the lower limit of the 95% CI is $> 50\%$, see Table x), suggesting there may be able to separate certain pairs of naturalistic canopies to some degree (assuming no other information was available to the wasps).

naturalistic test vs control



Produce necessary tables to report from in-line and to include in MS.

Write results here: Evidence that wasps can discriminate naturalistic canopies from the test vs control analysis is also mixed.

Coming soon: 1. comparison of naturalistic stimuli versus artificial

Q3 test vs control old code. redundant , can be thrown away

Misc code that can likely be thrown away: